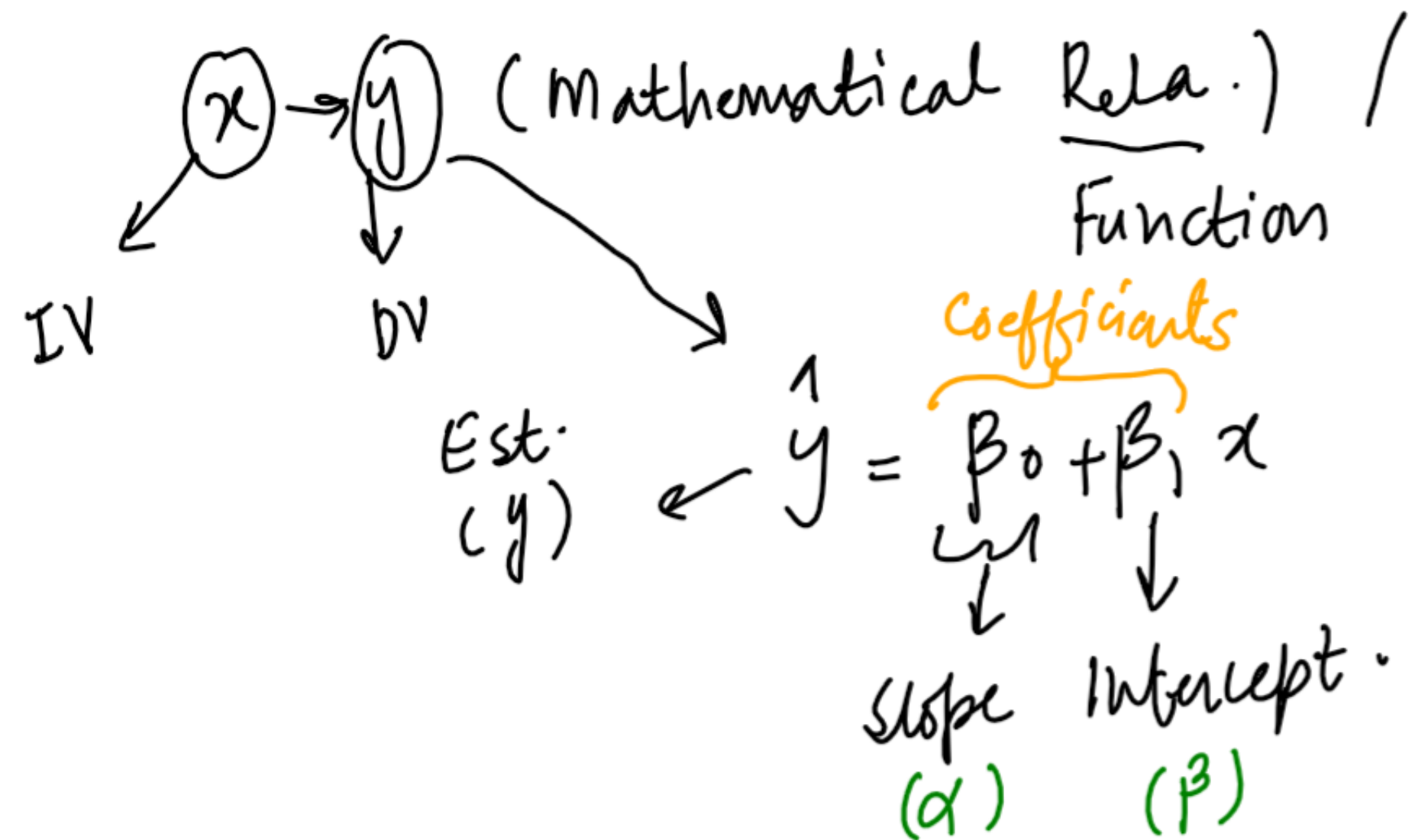


SIMPLE LINEAR REGRESSION: (Part-II)

$$\bar{x} = \frac{\sum x}{n}; \bar{y} = \frac{\sum y}{n}$$



$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n x_i^2 - \frac{(\sum x_i)^2}{n}}$$

$$\beta_0 = \frac{1}{n} \left(\sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i \right) = \bar{y} - \beta_1 \bar{x}$$

$$= \left(\frac{\sum y_i}{n} \right) - \beta_1 \left(\frac{\sum x_i}{n} \right)$$

$\bar{y}$ $\bar{x}$

- Mathematical function in modeling / forecasting.
Unknown / new variable info

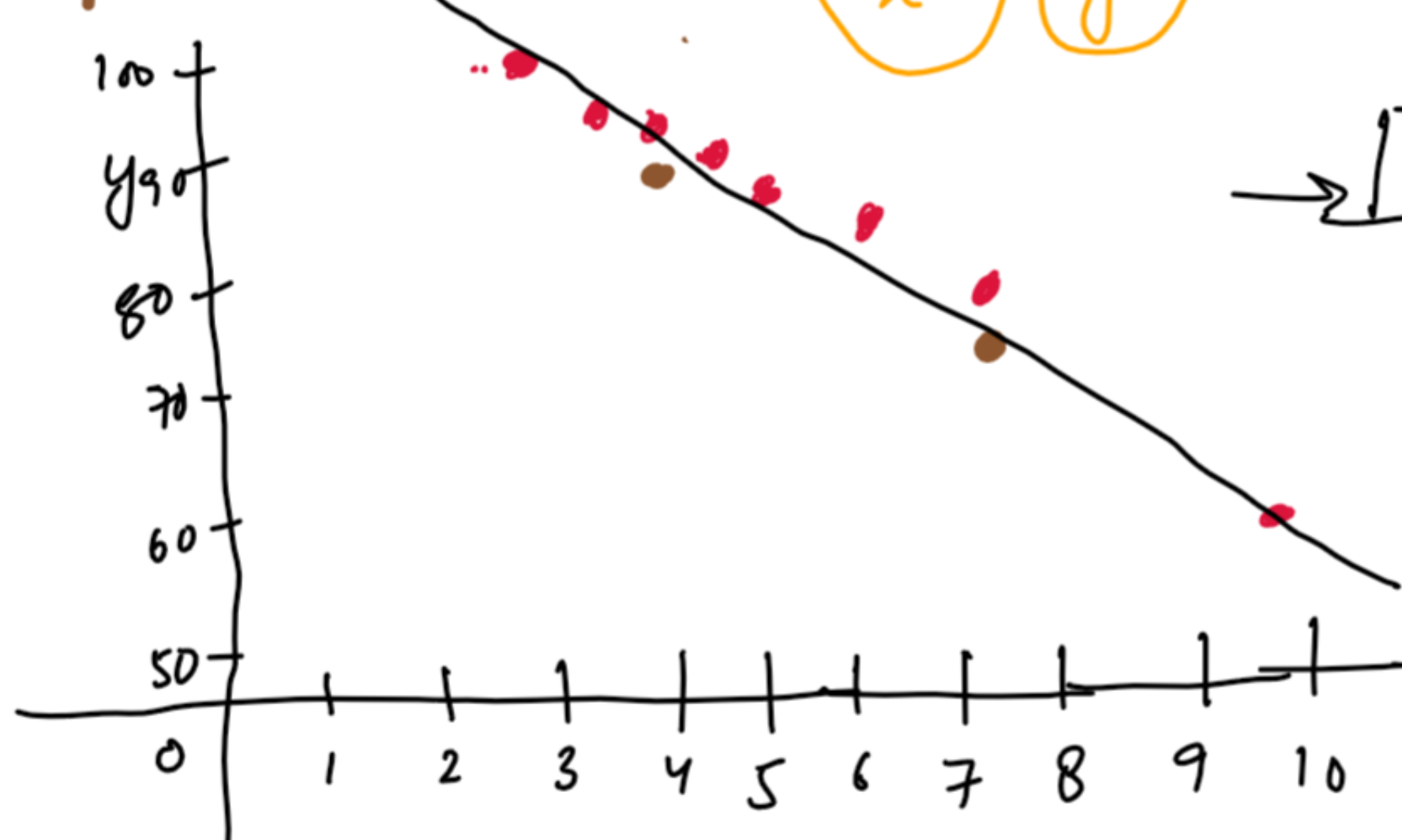
PHENOMENA MUST BE LINEAR

Exp: 1

14.98 Tax Efficiency. Following are the data on percentage of investments in energy securities and tax efficiency from Exercise 14.58.

Invest.	3.1	3.2	3.7	4.3	4.0	5.5	6.7	7.4	7.4	10.6
eff. ying.	98.1	94.7	92.0	89.8	87.5	85.0	82.0	77.8	72.1	53.5

DV / OUTCOME



PRIOR INFORMATION

LINEARITY

- 1 - 100
 2 - 98
 3 - 97
 4 - 60
 5 - 190
 116 $\rightarrow$ 18% GST.

14.98 Tax Efficiency. Following are the data on percentage of investments in energy securities and tax efficiency from Exercise 14.58.

x	3.1	3.2	3.7	4.3	4.0	5.5	6.7	7.4	7.4	10.6
y	98.1	94.7	92.0	89.8	87.5	85.0	82.0	77.8	72.1	53.5

n = 10

x	y
3.1	98.1
3.2	94.7
3.7	92.0
4.3	89.8
4.0	87.5
5.5	85.0
6.7	
7.4	
7.4	
10.6	

$$\sum x = 55.9; \sum y = 832.5$$

$$\sum x = 55.9$$

$$\sum y = 832.5$$

$$\sum xy = 4476.95$$

$$\sum x^2 = 365.05$$

$$\hat{y} = \beta_0 + \beta_1 x$$

$$= 102.03 + (-3.36)x$$

$$\hat{y} = 102.03 - 3.36x$$

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}$$

$$= \frac{\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}$$

$$\beta_0 = \frac{1}{n} \left(\sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i \right) = \bar{y} - \beta_1 \bar{x}$$

$$\beta_0 = \frac{\sum y_i}{n} - \beta_1 \frac{\sum x_i}{n}$$

$$\beta_1 = \frac{4476.95 - \frac{(55.9)(832.5)}{10}}{(365.05) - 312.48}$$

$$\beta_1 = \frac{4476.95 - 4653.67}{(365.05) - (312.48)} = \frac{-176.72}{52.57}$$

$$\beta_1 = -3.36$$

$$\beta_0 = 83.25 - (-3.36)(5.59)$$

$$= 83.25 + 18.78$$

$$= 102.03$$

$$(55.9)^2 = 3124.81$$